

C03_AT3

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1) "1. FIR Quantization using Truncation

Simulate a 16-tap FIR low-pass filter with cutoff frequency 0.4π rad/sample using 8-bit fixed-point coefficient truncation and analyze the resulting quantization error. (25) (BL3) (WK4)

Given Data:

- FIR Filter Length = 16 taps
- Cutoff Frequency = 0.4π rad/sample
- Coefficient Word Length = 8 bits
- Input Signal = Sum of 500 Hz and 2 kHz sinusoids
- Sampling Frequency = 8 kHz

Tasks:

1. Design the ideal FIR low-pass filter.
2. Quantize the coefficients using truncation.
3. Simulate the filter response using MATLAB/Python.
4. Plot the magnitude responses before and after quantization.
5. Determine the output error signal.
6. Calculate Mean Square Error (MSE)

CODE

```
%% =====  
% FIR Quantization using Truncation - 16-tap Low Pass Filter  
% -----  
% Filter Length    : 16 taps  
% Cutoff Frequency : 0.4*pi rad/sample  
% Coefficient Word Length : 8 bits (Q1.7 signed fixed-point)  
% Input Signal     : 500 Hz + 2 kHz sinusoids  
% Sampling Frequency : 8 kHz  
% =====
```

```
clear; clc; close all;
```

```
%% ----- Given Parameters -----
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N      = 16;      % Number of taps (filter length)
wc      = 0.4*pi;  % Cutoff frequency (rad/sample)
wc_norm = wc/pi;   % Normalized cutoff for fir1 (1 = Nyquist = pi)
bits    = 8;      % Coefficient word length
Fs      = 8000;    % Sampling frequency (Hz)
f1      = 500;     % Tone 1 (Hz)
f2      = 2000;    % Tone 2 (Hz)
N_samples = 500;   % Number of input samples
```

```
%% ----- Task 1: Design Ideal FIR Low-Pass Filter -----
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% Order = N-1 taps -> gives N = 16 coefficients, Hamming-windowed design
h_ideal = fir1(N-1, wc_norm, 'low', hamming(N));
```

```
fprintf('---- Ideal (Double Precision) Filter Coefficients ----\n');
disp(h_ideal);
```

```
%% ----- Task 2: Quantize Coefficients using Truncation -----
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% Assumption: 8-bit signed fixed-point, Q1.7 format
% (1 sign bit + 7 fractional bits), since |h_ideal| < 1
qstep = 2^-(bits-1);      % Quantization step size
```

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% Truncation (round toward -infinity -> two's complement truncation)
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h_quant = floor(h_ideal/qstep) * qstep;
```

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% Saturate to representable Q1.7 range [-1, 1-qstep]
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max_val = 1 - qstep;
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min_val = -1;
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```
h_quant = max(min(h_quant, max_val), min_val);
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```

fprintf('\n---- Quantized (8-bit Truncated) Filter Coefficients ----\n');
disp(h_quant);

%% ----- Task 3: Simulate Filter Response -----
n = 0:N_samples-1;
t = n/Fs;
x = sin(2*pi*f1*t) + sin(2*pi*f2*t); % Input = 500 Hz + 2 kHz sinusoids

y_ideal = filter(h_ideal, 1, x); % Output using ideal (double) filter
y_quant = filter(h_quant, 1, x); % Output using quantized filter

%% ----- Task 4: Magnitude Response Before & After Quantization -----
[H_ideal, w1] = freqz(h_ideal, 1, 1024, Fs);
[H_quant, w2] = freqz(h_quant, 1, 1024, Fs);

figure('Name','Magnitude Response');
plot(w1, 20*log10(abs(H_ideal)), 'b', 'LineWidth', 1.5); hold on;
plot(w2, 20*log10(abs(H_quant)), 'r--', 'LineWidth', 1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response: Ideal vs 8-bit Truncated FIR Filter');
legend('Ideal (Double Precision)', 'Quantized (8-bit Truncation)');

%% ----- Task 5: Output Error Signal -----
error_signal = y_ideal - y_quant;

figure('Name','Time Domain Outputs and Error');
subplot(3,1,1);
plot(t, y_ideal, 'b'); grid on;

```

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title('Filter Output - Ideal Coefficients');
xlabel('Time (s)'); ylabel('Amplitude');

subplot(3,1,2);
plot(t, y_quant, 'r'); grid on;
title('Filter Output - Quantized Coefficients (8-bit Truncation)');
xlabel('Time (s)'); ylabel('Amplitude');

subplot(3,1,3);
plot(t, error_signal, 'k'); grid on;
title('Output Error Signal (Ideal - Quantized)');
xlabel('Time (s)'); ylabel('Error Amplitude');

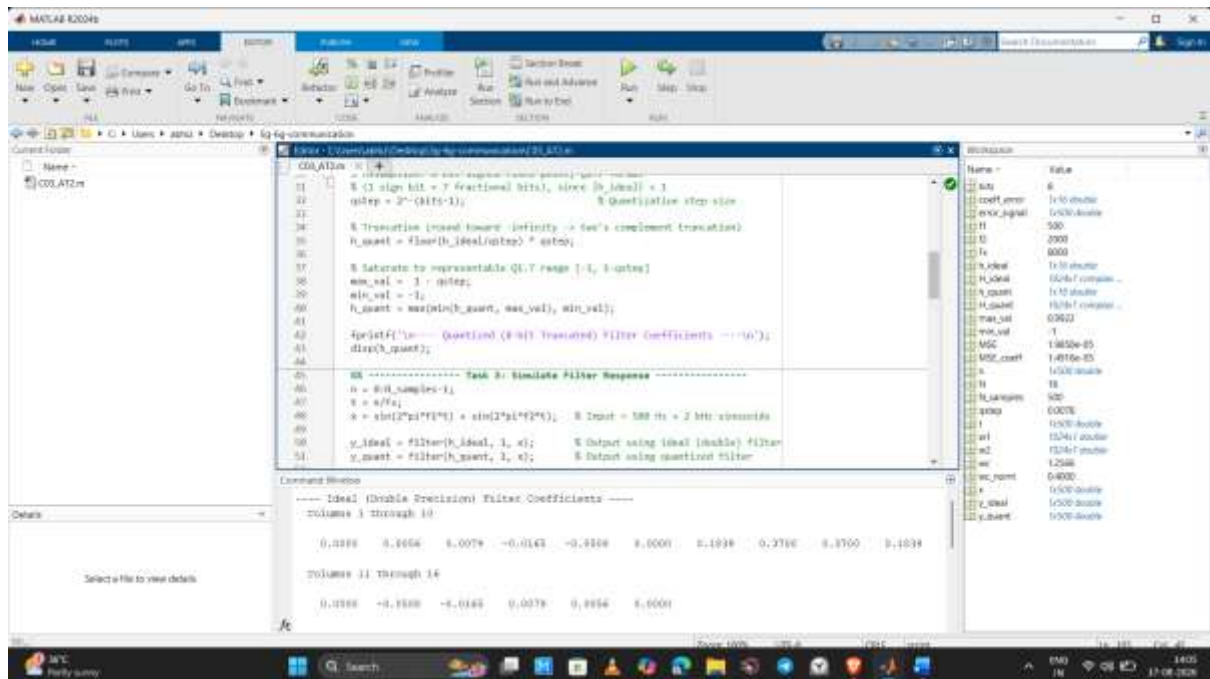
%% ----- Task 6: Mean Square Error (MSE) -----
MSE = mean(error_signal.^2);
fprintf('\n---- Performance Metrics ----\n');
fprintf('MSE between ideal and quantized filter OUTPUT = %e\n', MSE);

% Optional: coefficient-level quantization error
coeff_error = h_ideal - h_quant;
MSE_coeff = mean(coeff_error.^2);
fprintf('MSE of QUANTIZED COEFFICIENTS (vs ideal) = %e\n', MSE_coeff);

%% ----- Optional: Coefficient Comparison Plot -----
figure('Name', 'Coefficient Comparison');
stem(0:N-1, h_ideal, 'b', 'filled'); hold on;
stem(0:N-1, h_quant, 'r');
grid on;
xlabel('Tap Index (n)');
ylabel('Coefficient Value');
title('FIR Coefficients: Ideal vs 8-bit Truncated');

```

legend('Ideal', 'Quantized (Truncated)');



OUTPUT

---- Ideal (Double Precision) Filter Coefficients ----

Columns 1 through 10

0.0000 0.0056 0.0079 -0.0165 -0.0508 0.0000 0.1839 0.3700 0.3700 0.1839

Columns 11 through 16

0.0000 -0.0508 -0.0165 0.0079 0.0056 0.0000

---- Quantized (8-bit Truncated) Filter Coefficients ----

Columns 1 through 10

0 0 0.0078 -0.0234 -0.0547 0 0.1797 0.3672 0.3672 0.1797

Columns 11 through 16

```
0 -0.0547 -0.0234 0.0078 0 0
```

---- Performance Metrics ----

MSE between ideal and quantized filter OUTPUT = 1.984955e-05

MSE of QUANTIZED COEFFICIENTS (vs ideal) = 1.491620e-05

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